

torch.flatten#

<https://pytorch.org/docs/stable/generated/torch.flatten.html>

Description:

Flattens input by reshaping it into a one-dimensional tensor. If `start_dim` or `end_dim` are passed, only dimensions starting with `start_dim` and ending with `end_dim` are flattened. The order of elements in input is unchanged. Unlike NumPy's `flatten`, which always copies input's data, this function may return the original object, a view, or copy. If no dimensions are flattened, then the original object input is returned. Otherwise, if input can be viewed as the flattened shape, then that view is returned. Finally, only if the input cannot be viewed as the flattened shape is input's data copied. See `torch.Tensor.view()` for details on when a view will be returned. Flattening a zero-dimensional tensor will return a one-dimensional view. `input` (Tensor) – the input tensor.

Function Signature

```
torch.flatten(input, start_dim=0, end_dim=-1) → Tensor#
```

Parameters

Code Examples

```
>>> t = torch.tensor([[[1, 2],
...                    [3, 4]],
...                  [[5, 6],
...                  [7, 8]]])
>>> torch.flatten(t)
tensor([1, 2, 3, 4, 5, 6, 7, 8])
>>> torch.flatten(t, start_dim=1)
tensor([[1, 2, 3, 4],
        [5, 6, 7, 8]])
```

Notes & Warnings

NOTE Note Flattening a zero-dimensional tensor will return a one-dimensional view.

Detailed Documentation

Documentation

Flattens input by reshaping it into a one-dimensional tensor. If `start_dim` or `end_dim` are passed, only dimensions starting with `start_dim` and ending with `end_dim` are flattened. The order of elements in input is unchanged.

Unlike NumPy's `flatten`, which always copies input's data, this function may return the original object, a view, or copy. If no dimensions are flattened, then the original object input is returned. Otherwise, if input can be viewed as the flattened shape, then that view is returned. Finally, only if the input cannot be viewed as the flattened shape is input's data copied. See `torch.Tensor.view()` for details on when a view will be returned.

Flattening a zero-dimensional tensor will return a one-dimensional view.

`input` (Tensor) – the input tensor.

`start_dim` (int) – the first dim to flatten

`end_dim` (int) – the last dim to flatten